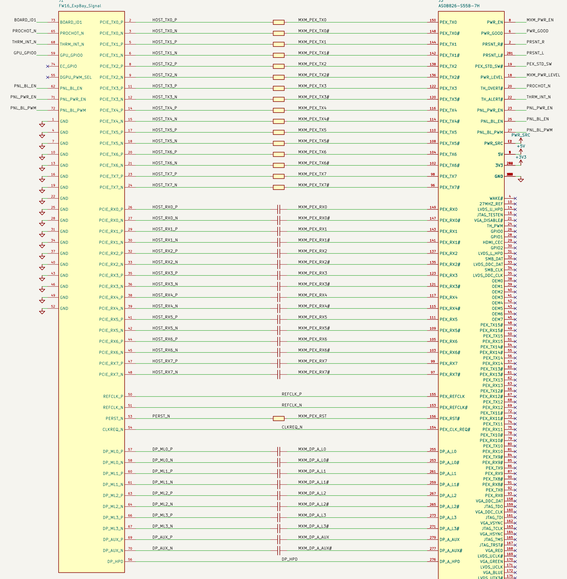
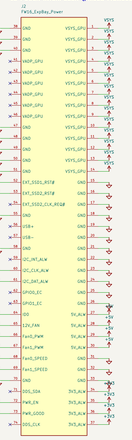
FW16 EXPANSION BAY -> MXM 3.x ADAPTER | schematic v4 DRAFT | generated + ERC-checked with KiCad 10 (kicad-cli)

NOT A MANUFACTURABLE DESIGN. No PCB layout. Every value/pin/footprint marked TBD is an unresolved placeholder (see BOM CSV).
J1/J2 pins: github.com/FrameworkComputer/ExpansionBay Electrical/README.md ; J3: MXM 3.1 connector tables (ADLINK MXM-AXe, Advantech SKY-MXM-RTX3000).
Lanes go straight through by name: host PCIe_TX -> MXM PEX_TX (MXM names are system-relative; confirmed by MXM 3.1 Table 3.6).
Series parts: host->card PCIe R20-R35 = 0R; card->host PCIe C10-C25 = 220nF; DP_A lanes+AUX C26-C35 = 100nF DC blocking; HPD direct (R14 pull-down DNP).
OPEN: TH_PWM unconnected. J3 = 555B on a 1.2 mm adapter; needs the flat 30-printed bay cover (15.3 mm inside). Pins 52/53 and the F75303 address per FW16 7040 interface schematic. Checked against MXM 3.1 r1.0.



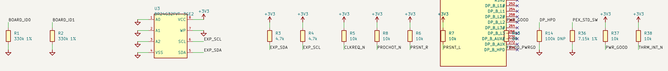
J2 POWER INTERPOSER (pinout: FrameworkComputer/ExpansionBay)



FAN HEADERS — host EC drives 12V_FAN / PWM (5 V) and reads SPEED (5 V OD). Pinout per FW ref PCB drawing.



EC MODULE-DETECTION + PULL-UPS (330k IDs work per forum t/74018 OP; other values from MXM 3.1, the FW16 interface schematic and the community board)



EEPROM: blank = EC defaults to dual-SSD (x4x2) and never enables VSYS. Program PCIE_THERMAL_3RDPARTY (PCIE_RX1, VSYS_EN, FAN_EN, 3V_5V_EN, F75303 @0x40).
but set GPU_PCIE_MUX_SEL LOW so DP_A AUX is not rerouted to SSD2 CLK. Note: EC sets GPU_PRESENT (eDP mux path) only for vendor GPU_AMD_R23M.

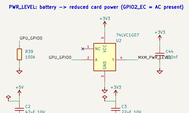
FAN-CONTROL TEMP SENSOR — F75303 @0x40 on the ALW I2C bus (Framework PCIE_THERMAL_3RDPARTY descriptor). Pins per FW16 7040 interface schematic p.9.



POWER: VSYS -> TVS D1 -> eFuse U1 (TPS25982, ~9.9 A limit) -> PWR_SRC to the MXM card; +5V / +3V3 from host 5V_ALW (2.5A max) / 3V3_ALW (2.0A max)



PWR_LEVEL: battery -> reduced card power (GPIO2_EC = AC present)



PWR_EN: 1.8 V host GPIO AND eFuse PG -> 3.3 V card PWR_EN



ERC + netlist checked with KiCad 10 kicad-cli. Not validated electrically. TBD = unresolved.

Sheet: /
File: fw16_mxm_adapter.kicad_sch

Title: Framework 16 Expansion Bay to MXM 3.x Adapter

Size: A3 | Date: 2026-11-08 | Rev: v4-DRAFT
KiCad E.D.A. 10.0.0 | ID: 1/1